Homework 25: 10.1-10.3 **Name** _____
AP Calculus BC **Due** _____
Period _____ **Score** _____ **/ 100**

Skill Builder 10.1 – Defining Convergent and Divergent Infinite Series

Write the first 5 terms of the sequence.

1.) $\{a_n\} = \left\{\dfrac{2n}{n+2}\right\}$	**2.)** $\{a_n\} = \{\cos n\pi\}$
3.) $\{a_n\} = \left\{(-1)^n\left(\dfrac{1}{n}+\dfrac{1}{n^2}\right)\right\}$	**4.)** $\{a_n\} = \left\{\dfrac{(2n)!}{(n+1)!}\right\}$

Write the first 5 terms of the sequence.

5.) $a_1 = 3,\ a_n = a_{n-1} + 1$	**6.)** $a_1 = 2,\ a_n = (a_{n-1})^2 - 4$
7.) $a_1 = 1,\ a_n = 3(a_{n-1}) - 2n$	**8.)** $a_1 = 4,\ a_n = ((a_{n-1}) - n)^2$

Write a non-recursive expression for the *n*th term of the sequence.

9.) $\{1, 5, 9, 13, \ldots\}$	**10.)** $\left\{\dfrac{1}{2}, \dfrac{1}{3}, \dfrac{1}{4}, \dfrac{1}{5}, \ldots\right\}$

11.) $\left\{2, -1, \frac{1}{2}, -\frac{1}{4}, \frac{1}{8}, \ldots\right\}$	**12.)** $\{-1, 2, 7, 14, 23, \ldots\}$

In the following sequences, determine the convergence or divergence with the given *n*th term. Be sure to properly justify your conclusion.

13.) $\{a_n\} = \left\{(-1)^n \left(\frac{n}{n+1}\right)\right\}$	**14.)** $\{a_n\} = \{n \pm (-1)^n\}$
15.) $\{a_n\} = \left\{\frac{3n^2 - 3n + 8}{4n^2 + 1}\right\}$	**16.)** $\{a_n\} = \left\{\frac{-2\sqrt{n}}{3\sqrt{n} + 1}\right\}$
17.) $\{a_n\} = \left\{\frac{\ln(n^3 + 1)}{n}\right\}$	**18.)** $\{a_n\} = \left\{\frac{4^n}{5^n}\right\}$

19.) $\{a_n\} = \left\{\dfrac{(n+2)!}{(n+1)!}\right\}$	**20.)** $\{a_n\} = \left\{\dfrac{n!}{(n+2)!}\right\}$
21.) $a_n = n\sin\dfrac{1}{n}$	

Find the sequence of partial sums, S_1, S_2, S_3, S_4.

22.) $1+\dfrac{1}{4}+\dfrac{1}{9}+\dfrac{1}{16}+\dfrac{1}{25}+\cdots$	**23.)** $\displaystyle\sum_{n=1}^{\infty}\frac{(-1)^{n+1}}{n!}$

Skill Builder 10.2 – Working With Geometric Series

In the following exercises, find the sum of the convergent geometric series. Be sure to properly justify your conclusion.

1.) $\sum_{n=0}^{\infty}\left(\frac{1}{3}\right)^n$	**2.)** $\sum_{n=0}^{\infty}2\left(\frac{4}{5}\right)^n$
3.) $\sum_{n=0}^{\infty}\left(-\frac{1}{2}\right)^n$	**4.)** $40+16+\frac{32}{5}+\frac{64}{25}+\cdots$
5.) $60-30+15-\frac{15}{2}+\cdots$	**6.)** $-55+5.5-0.55+0.055-\cdots$

In the following exercises, find the interval that expresses all the x values for which each of the given series converges.

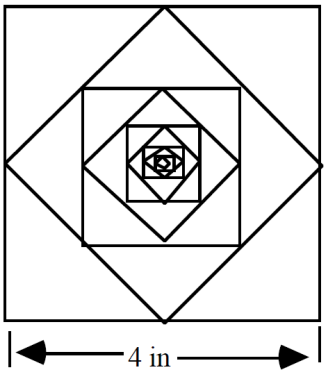
7.) $\sum_{n=1}^{\infty}3\left(\frac{x}{5}\right)^n$	**8.)** $\sum_{n=1}^{\infty}2\left(\frac{x-7}{3}\right)^n$

9.) Jack is sick and takes a flu medication every hour. He takes 100 mg initially and each dosage drops by 20%. If he keeps taking the medication forever, how much will he have taken? Jill is also sick. Each dosage of hers drops by 30%. If she ends up taking half the amount of medication that Jack eventually took, what was her original dosage?

10.) The annual spending by tourists in a resort city is $100 million. About 75% of that revenue is spent again in the resort city, and of that amount, about 75% is again spent in that city and so on. Write the geometric series that gives the total amount of spending generated by the $100 million and the sum of that series.

11.) This is a square of side 4 inches. The midpoints of the square are the vertices of an inscribed square. More squares are inscribed using the same pattern forever. The perimeters of the squares represents a geometric sequences.

a. Determine if the infinite sum of these perimeters converges and if so, find the sum.



b. The sum of the area of the squares is also a geometric series. Does this series converge or diverge? If it converges, to what value does it converge?

In the following exercises, (a) write the repeating decimal as an infinite geometric series and (b) express the sum of the series as a reduced fraction.

12.) $0.\overline{4}$	**13.)** $0.\overline{81}$

In the following exercises, determine the sum of the convergent series. Be sure to properly justify your conclusion.

14.) $\sum_{n=1}^{\infty}\left(\frac{1}{n}-\frac{1}{n+2}\right)$

15.) $\sum_{n=1}^{\infty}\frac{4}{(n+2)(n+4)}$

16.) $\sum_{n=1}^{\infty}\frac{4}{n(n+2)}$

Skill Builder 10.3 – The *n*th Term Test for Divergence

In the following exercises, determine the convergence or divergence of the series. Be sure to properly justify your conclusion. Not all series will require the use of the *n*th Term Test for Divergence.

1.) $\sum_{n=1}^{\infty} \frac{n+5}{5n+2}$	**2.)** $\sum_{n=1}^{\infty} \frac{n+3}{2n-1}$
3.) $\sum_{n=1}^{\infty} (\sin 1)^n$	**4.)** $\sum_{n=1}^{\infty} \frac{2^n}{n^2}$
5.) $\sum_{n=0}^{\infty} \frac{9}{3^n}$	**6.)** $\sum_{n=2}^{\infty} \frac{n}{10 \ln n}$